PRACTICAL-5

1. Aim

Interface ADS 1115 with Raspberry Pi to act as oscilloscope.

2. Prior Concepts

i) Concept of Python Programming

ii) Use of Gas Sensor

ii) Concept of Oscilloscope

3. New Concepts

i) Hardware interfacing of Gas Sensor with ADS1115 ADC and Raspi.

ii) Generating waveform of Gas Sensor values.

4. Hardware Requirement and PIN Connection

i) Raspberry Pi 4

ii)ADS1115 ADC

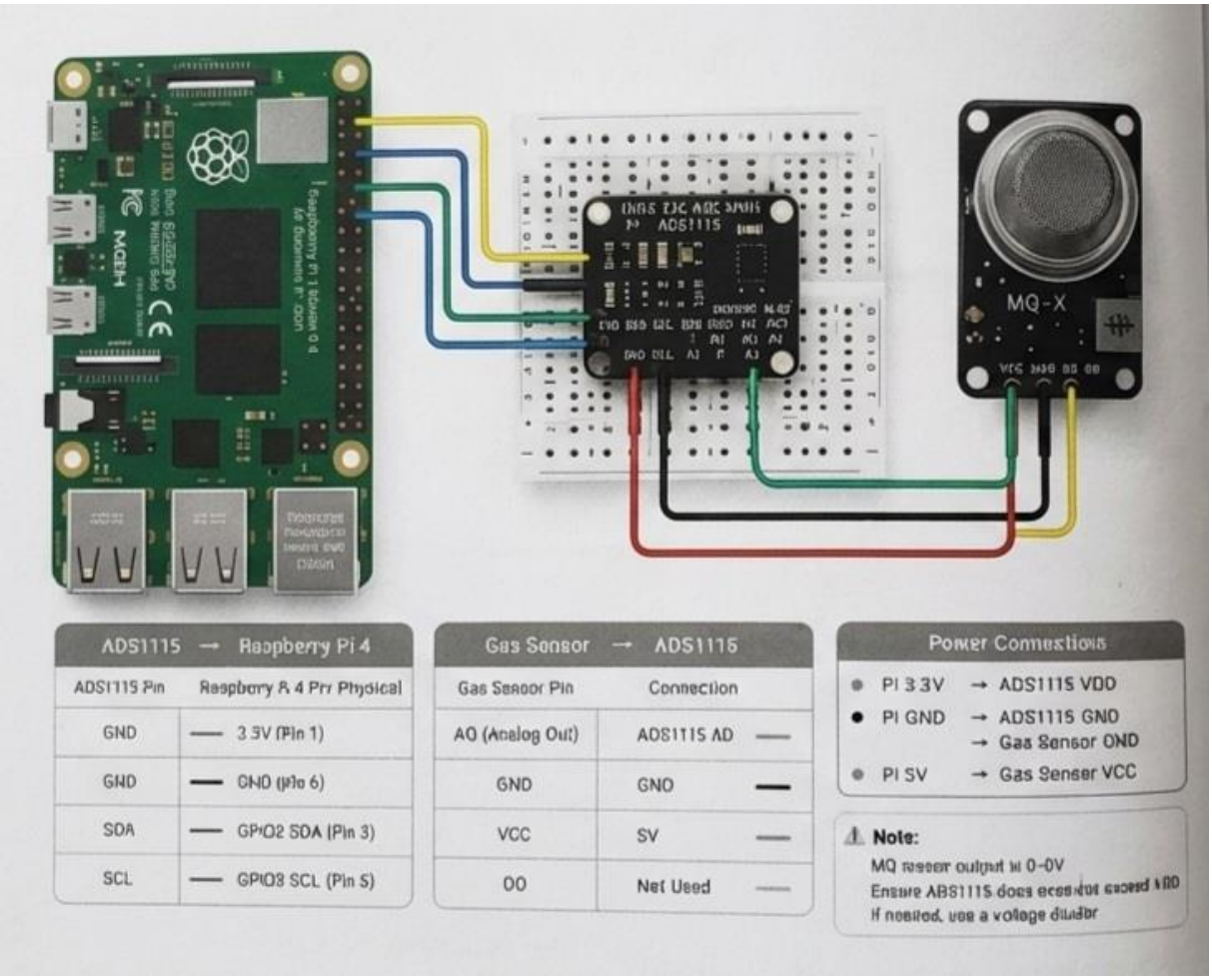
iii) MQ-2 Gas Sensor

iv)Jumper wires

Pin Connection

ADS1115	RPi Pin	Function
VDD(5V)	2	5V
GND	6	GND(GROUND)
SDA	3	GPIO 2(SDA1)
SCL	5	GPIO 3(SCL1)

MQ2 Gas Sensor	ADS1115	RPi Pin	Function
VCC	-	4	5V
GND	-	39	GND(GROUND)
A0	A0	-	-



5. Procedure

Step 1: Enable Raspberry Pi I2C interface

To enable the I2C, from the terminal, run;

sudo raspi-config

When the configuration panels open, select interface options, select I2C and click enable.

Step 2: Update the Raspberry pi

To do this, run below two commands:

sudo apt update

Step 3: Install the Adafruit ADS1115 library for ADC and Matplotlib for plotting of graph.

pip3 install adafruit-circuitpython-ads1x15 matplotlib --break-system-packages

pip3 uninstall -y adafruit-platformdetect adafruit-blinka

sudo reboot

pip3 install --no-cache-dir --upgrade adafruit-platformdetect adafruit-blinka

sudo reboot

Note:

Once all the above configuration command run and connection done then below command should be run.

sudo apt-get install -y i2c-tools python3

Ismod | grep i2c_

i2cdetect -y 1

Write the below code:

```
import time
import board
import busio
import matplotlib.pyplot as plt
from collections import deque
import adafruit_ads1x15.ads1115 as ADS
from adafruit_ads1x15.analog_in import AnalogIn
i2c = busio.I2C(board.SCL, board.SDA)
ads = ADS.ADS1115(i2c)
ads.gain = 1 # ±4.096
chan = AnalogIn(ads, ADS. PO)
x_vals = deque(maxlen = 50)
Y_vals = deque(maxlen = 50)
```

```
pit.ion()
fig, ax = plt.subplots()
line, = ax.plot([],[], label="Gas Sensor Values from ADS1115", marker='o') ax.set_ylim(0,2)
ax.set_xlim(0,100)
ax.set_title('MQ2 Gas Sensor Live Plot')
ax.set_xlabel('Sample')
ax.set_ylabel('Voltage(V)')
ax.legend()

counter = 0

print("Reading Gas Sensor values in Volts...!")
try:
        while True:
        v = chan.voltage
        x_vals.append(counter)
        y_vals.append(v)
        line.set_xdata(x_vals)
        line.set_ydata(y_vals)
        ax.set_xlim(counter-100 if counter > 100 else 0, counter + 1)
        fig.canvas.draw()
        fig.canvas.flush_events()
        time.sleep(1)
        counter+=1

except KeyboardInterrupt:
print("Stopped")
```

6. Results: